STICS crop model general presentation

1. Introduction : History, purpose, structure and types of application.
2. Description of soil and formalization of energy, water, Nitrogen balance
3. Formalizations of development and growth of crops
4. Need in inputs and care for options
5. Principles for calibration and test

Principles for calibration and test

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ALIMENTATION
AGRICULTURE
ENVIRONNEMENT

foreword

Semantic:

- Parameterization = fixing all the inputs using literature, direct measurement or paraméter fitting
- Calibration = fixing global input
- Forcing (including assimilation) concerns output as opposed input.

Warning : A fitted parameter will both depend on

- database quality
- model performance for predicting all the variables which interact in the involved process

1) Model evaluation

Before saying: « the model fails » needs a formal evaluation as regard the model objectives.

2 ways of model evaluation (Loague K., Green R.E., 1991).

- Comparing Prediction against observed values
- Implementing Sensitivity study analysis

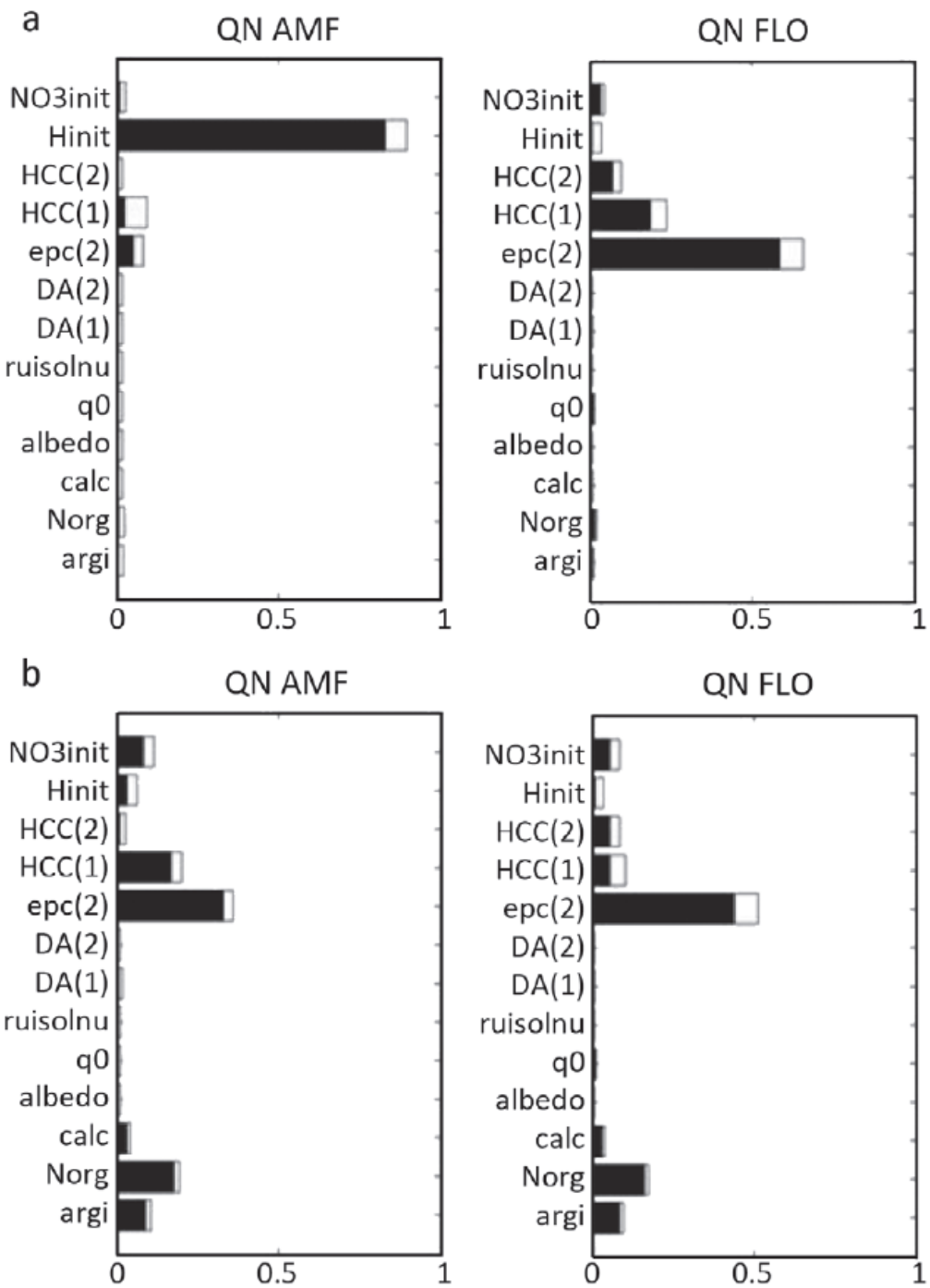


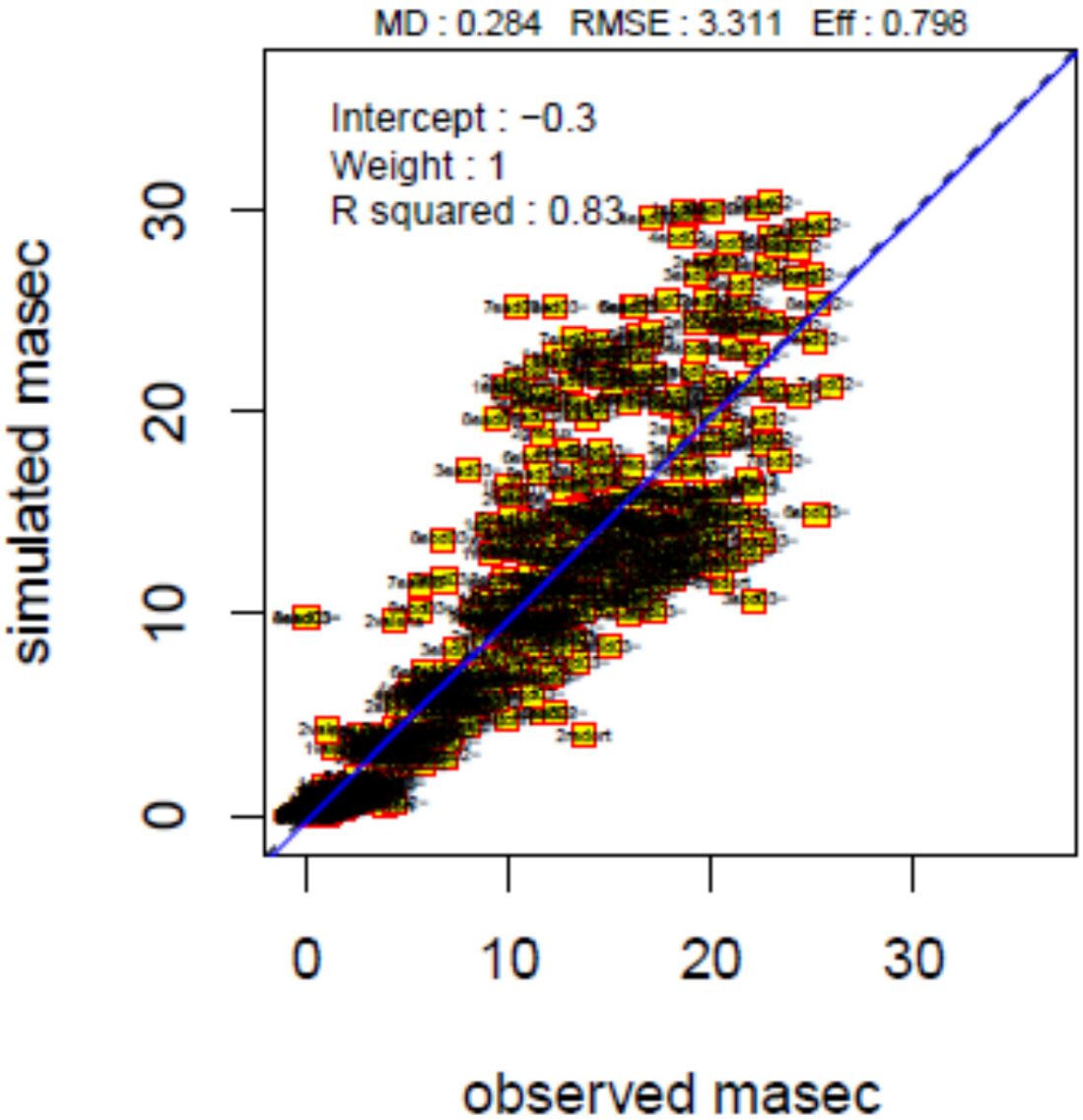
Fig. 14–6. Sensitivity indices of soil characteristics for the output variable aboveground plant N (QN) at beginning of stem elongation (AMF) and at flowering (FLO) for (a) dry climate and (b) wet climate. The first-order effects are represented in dark and interaction effects in light.

Sensitivity study

(Buis et al. 2011)

=> Be careful to general conditions and term of the sensitivity study

Timless comparing prediction vs observation



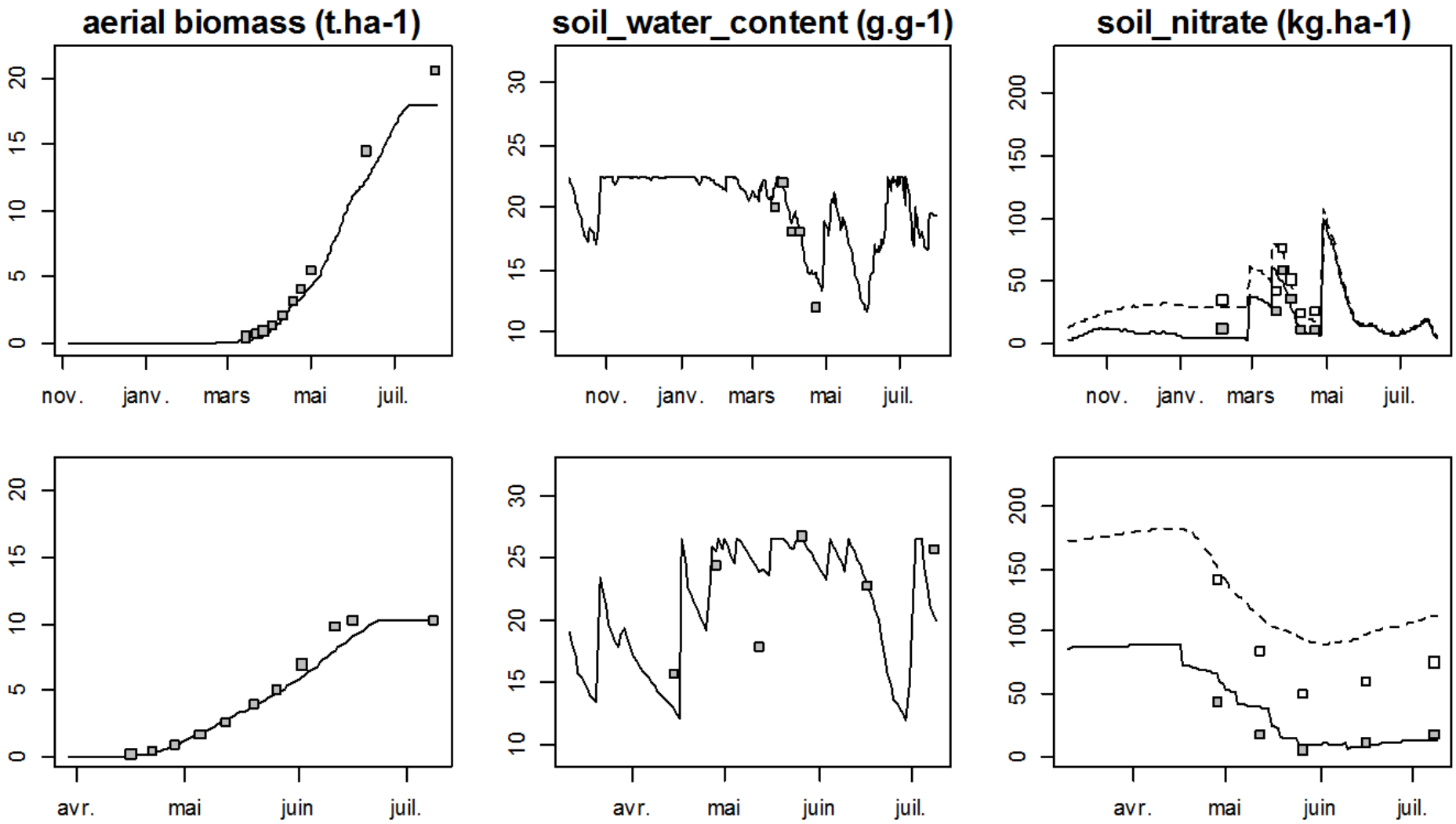
Take care to

- choice of variables of interest vs objectives
- Qatabase quality
- Complementary between visual andstatiscal evaluation

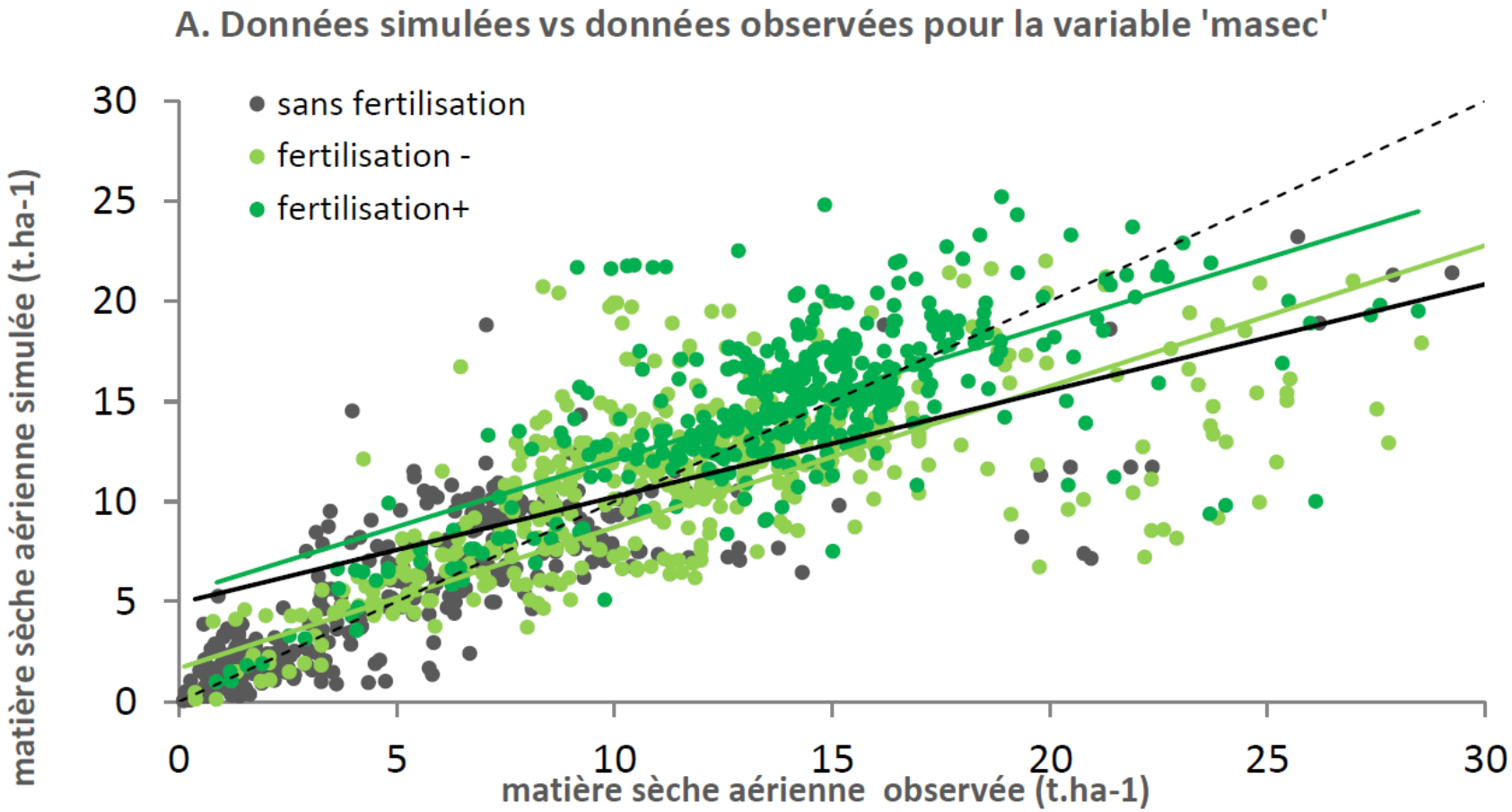
Statistical criteria => plural

		Definition
Least square regression		n being the number of observations O: observed value $<O>$: mean of the observed values P value predicted by STICS $\overline{P} = A \times O + B$: value predicted by least square regression
		slope intercept R2 = coefficient of determination
Root mean square error	RMSE	and (Relative value in %) $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(P_i - O_i)^2}$ and $\left(\frac{RMSE}{<O>}\right)$
	RMSEs	Systematic root mean square error (systematic part of RMSE in %) $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\overline{P}_i - O_i)^2}$ and $\left(\frac{RMSEs^2}{RMSE^2}\right)$
	RMSEu	unystematic root mean square error (unsystematic part of RMSE in %) $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(P_i - \overline{P}_i)^2}$ and $\left(\frac{RMSEu^2}{RMSE^2}\right)$
MD: mean deviation		$\frac{1}{n}\sum_{i=1}^{n}(P_i - O_i)$ and $\left(\frac{MD}{<O>}\right)$
EF: model efficiency		$1 - \frac{\sum_{i=1}^{n}(P_i - O_i)^2}{\sum_{i=1}^{n}(<O> - O_i)^2}$

Test using kinetics (Coucheney et al; in prep)

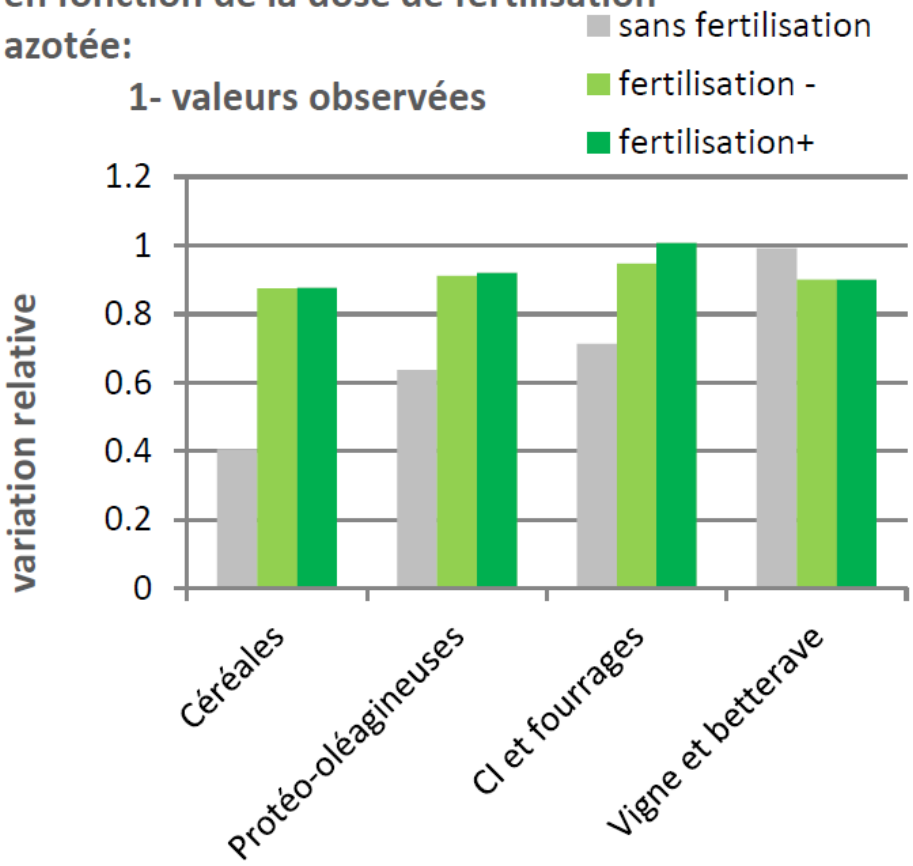


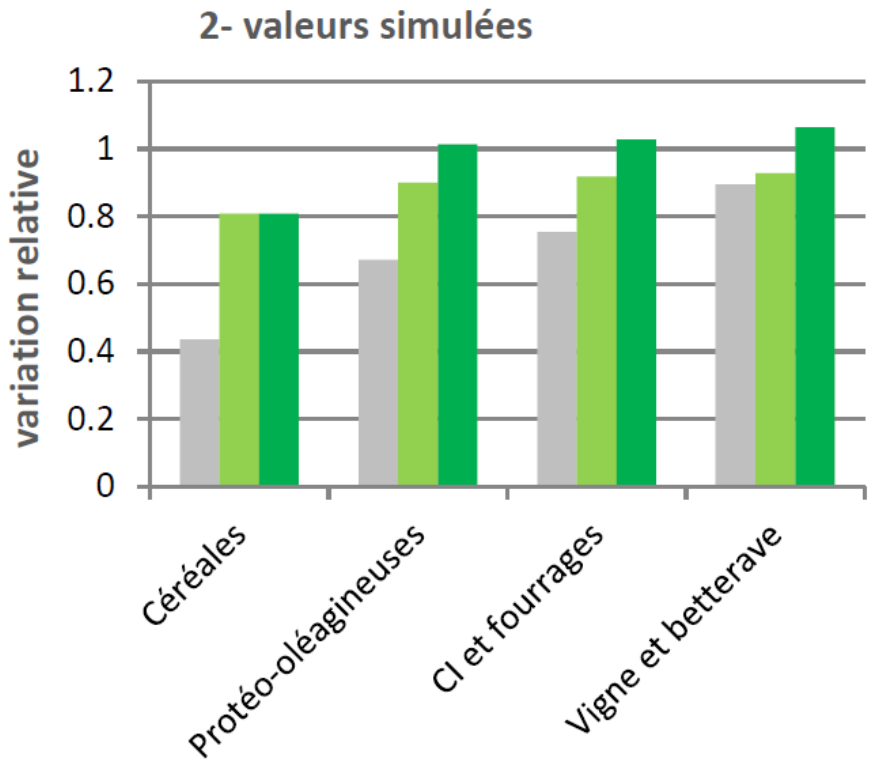
Test mixing comparison and sensitivity (Coucheney et al;



Comparison and sensitivity (Coucheney et al; 2012)

B. Variation relative de la variable 'masec' en fonction de la dose de fertilisation azotée:





How deals with local model discrepancy?

- Compare discrepancy with the published RMSE for similar conditions and in global tests.
- Check local model inputs.
- Implementing a simple sensitivity study analysis for some uncertain inputs (substitute file; change parameter value)
- Share problem with experts who can decide a formal recalibration of a global parameter
- If necessary, create an ad hoc model (Affolder et al., 2013)

Current approach for STICS calibration

1) Identify what parameter can be determine thanks to the scientific literature (the best way: Grant et al, 2003); possibility extrapolate form other STICS files (i.e similar crop)

2) Implement a sensitivity study analysis for some uncertain inputs => hierarchy of parameters to be calibrated (or see literature).

3) Gather and qualify dataset (an metadata) in term of representativity of conditions (potential, stressed) and available measured variable (espcially, LAI, roots,soil state…)

⇒ Share dataset in 2 parts : for calibration and test.

⇒ Check the current description of soil, climate, technics of data.

4) Calibrate some parameter using javaSTICS optimization or trial/error + R routine (see after) or Optimistics (Wallach & al.,2011).

5) Test against an extended dataset.

R routine : simulated vs observed comparison

Free loading at the STICS website

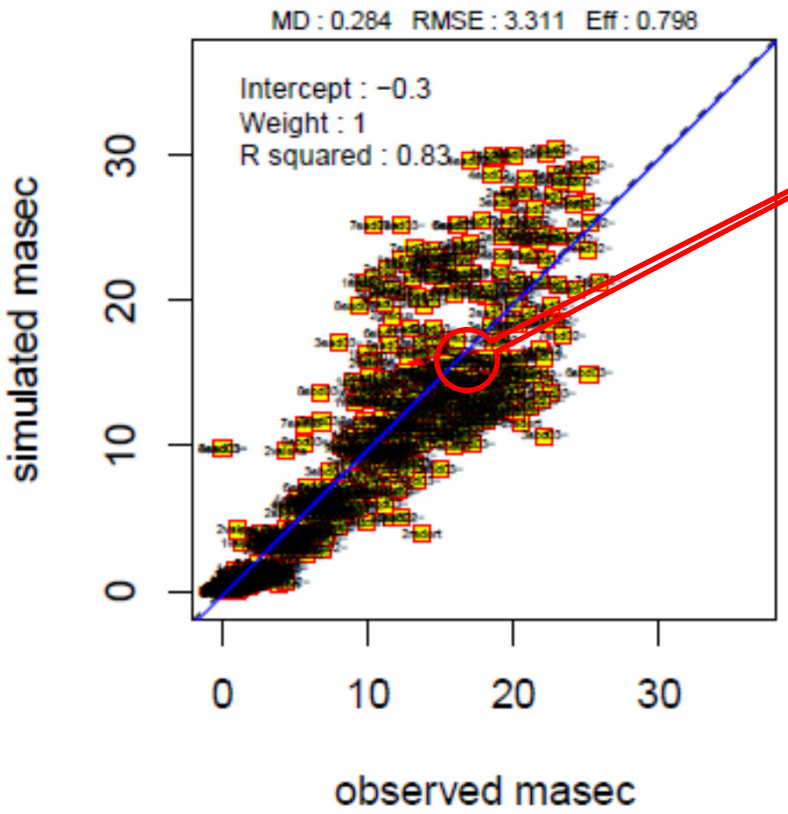
1) Collecting each (*s.sti and *.obs) file couple of files of all the usm of a directory

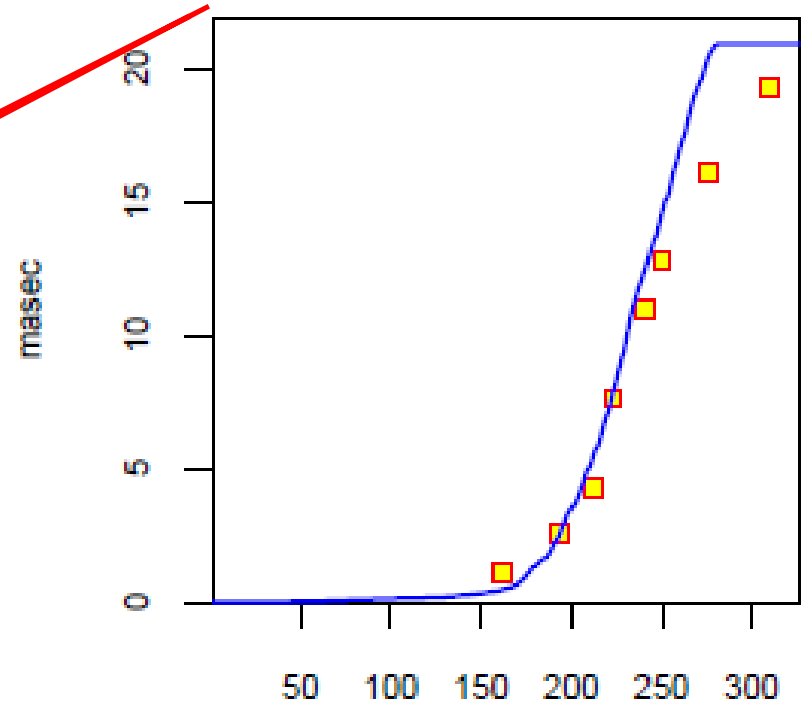
2) Building matrix and calculating statistics.

3) Building graphs (simulated vs observed or kinetics of simulated and observed.

Outputs of R routine

usm cn00_p5 variable: masec





Winter wheat : BD sms

The Modulo-Stics version

Thank you for your attention